

## Project Initialization and Planning Phase

|               |                                                |
|---------------|------------------------------------------------|
| Date          | 26 June 2026                                   |
| Team ID       |                                                |
| Project Name  | Smart Lender – Loan Approval Prediction System |
| Maximum Marks | 3 Marks                                        |

### Define Problem Statements (Customer Problem Statement Template):

The Smart Lender – Loan Approval Prediction System is designed to address the challenges faced by both loan applicants and financial institutions during the loan approval process. Traditional loan evaluation methods are often time-consuming, involve extensive manual verification, and may result in delays or inconsistencies in decision-making. Applicants are usually uncertain about their eligibility and have limited understanding of the factors influencing loan approval.

The objective of this project is to develop a machine learning-based prediction system that analyzes applicant details such as demographic information, financial status, credit history, loan amount, and property area to predict the likelihood of loan approval. By providing instant and reliable predictions through a user-friendly web application, the system improves transparency, reduces processing time, and supports financial institutions in making faster and more consistent lending decisions while helping applicants assess their eligibility before applying for a loan.

| <b>Problem Statement (PS)</b> | <b>I am (Customer)</b>                         | <b>I'm trying to</b>                                | <b>But</b>                                            | <b>Because</b>                | <b>Which makes me feel</b>            |
|-------------------------------|------------------------------------------------|-----------------------------------------------------|-------------------------------------------------------|-------------------------------|---------------------------------------|
| PS-1                          | A loan applicant seeking financial assistance. | Check my loan application is likely to be approved. | I don't know if I meet the bank eligibility criteria. | Many factors affect approval. | Uncertain about the result.           |
| PS-2                          | A home loan applicant.                         | Get a quick loan decision.                          | The process is slow.                                  | Banks verify many details..   | Frustrated by the delay.              |
| PS-3                          | A bank loan officer.                           | Evaluate applications quickly.                      | Many applications arrive daily.                       | Manual checks take time.      | Pressured to make accurate decisions. |